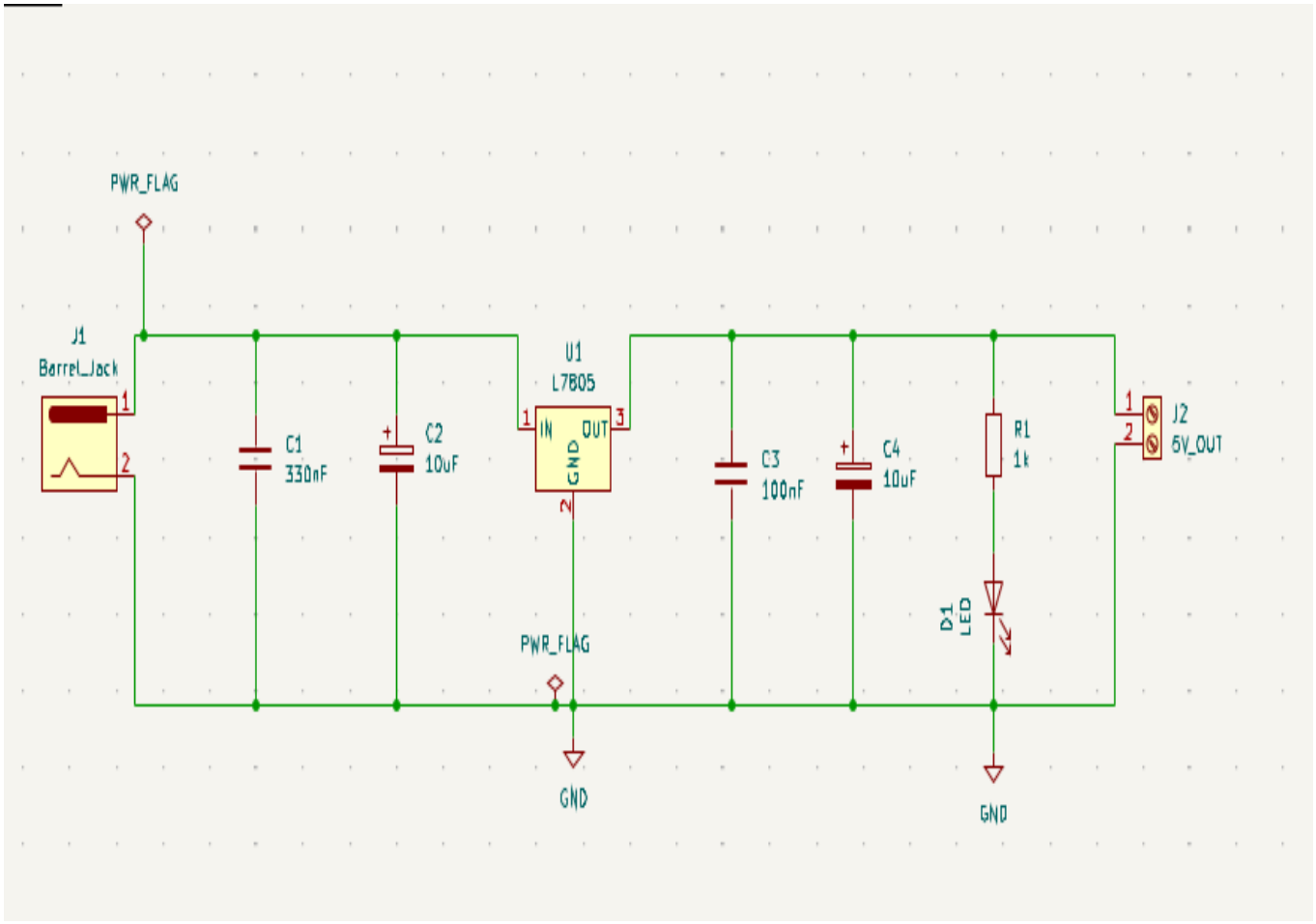
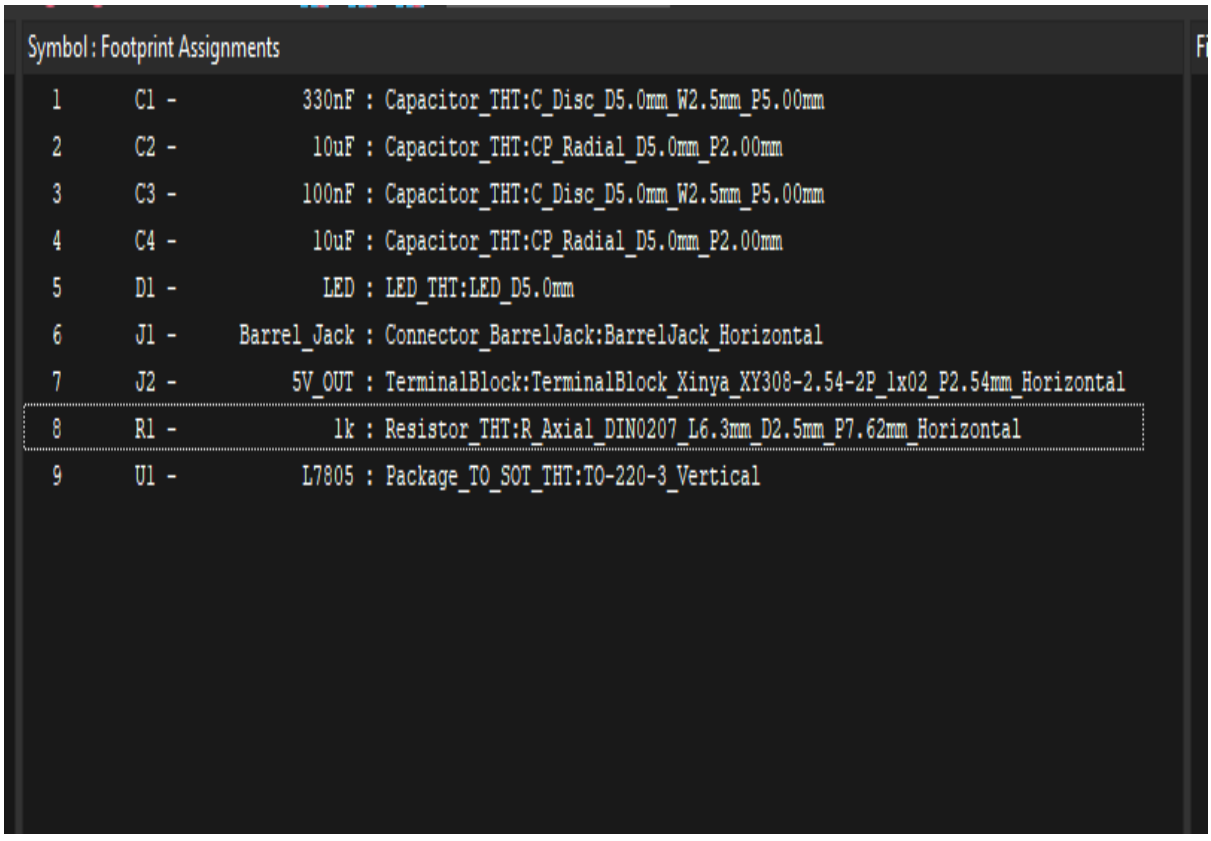
NAME: Tanmay Jamloki

Title : Power Supply Conditioning Board

3. Schematic



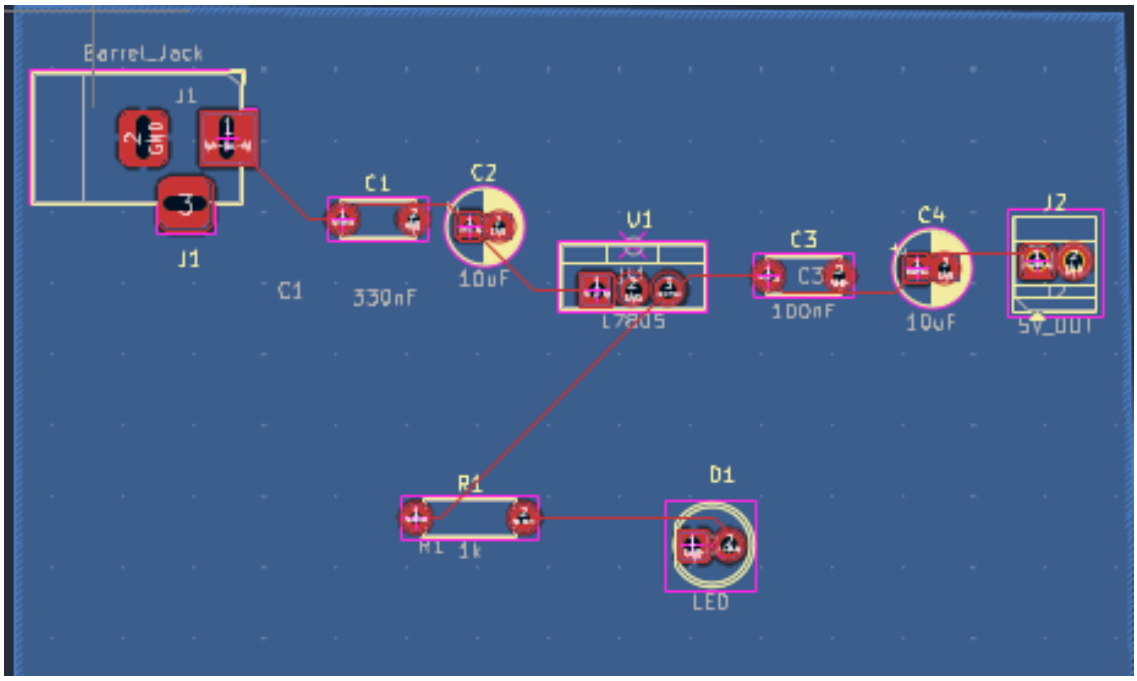
4. Screenshot of footprint after routing is completed



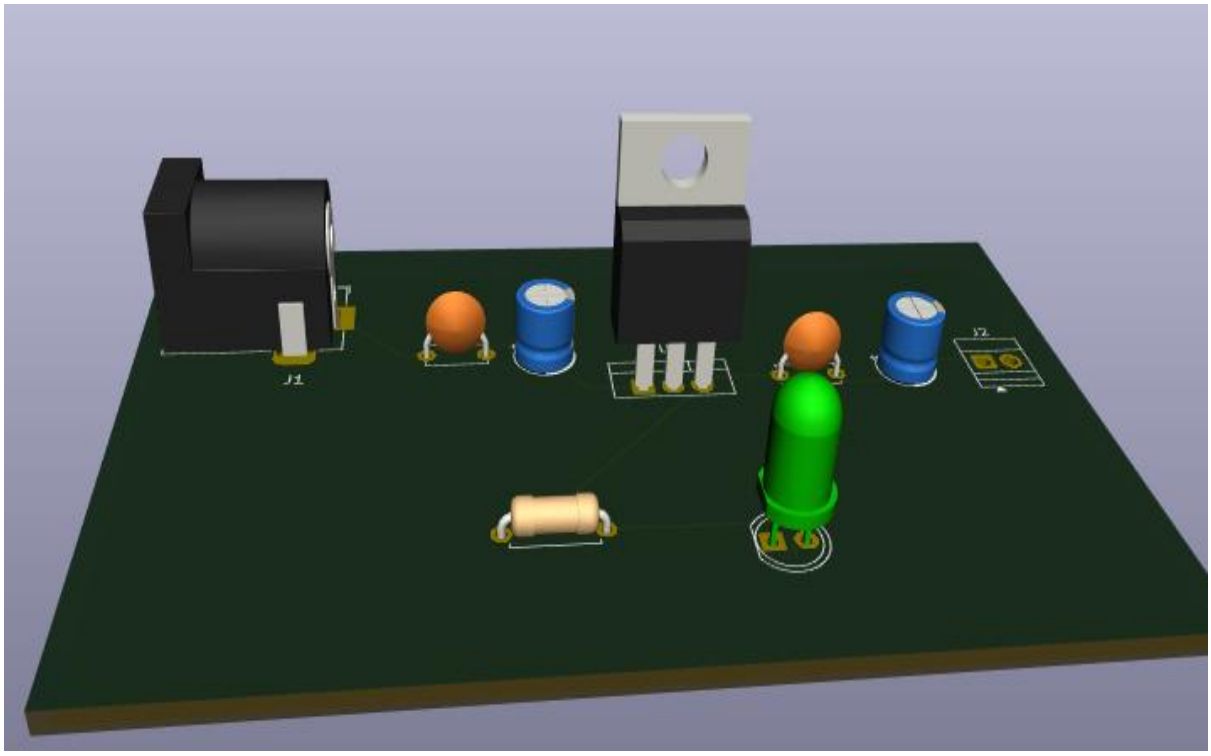
Symbol : Footprint Assignments

1	C1 -	330nF	Capacitor_THT:C_Disc_D5.0mm_W2.5mm_P5.00mm
2	C2 -	10uF	Capacitor_THT:CP_Radial_D5.0mm_P2.00mm
3	C3 -	100nF	Capacitor_THT:C_Disc_D5.0mm_W2.5mm_P5.00mm
4	C4 -	10uF	Capacitor_THT:CP_Radial_D5.0mm_P2.00mm
5	D1 -	LED	LED_THT:LED_D5.0mm
6	J1 -	Barrel_Jack	Connector_BarrelJack:BarrelJack_Horizontal
7	J2 -	5V_OUT	TerminalBlock:TerminalBlock_Xinya_XY308-2.54-2P_1x02_P2.54mm_Horizontal
8	R1 -	1k	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
9	U1 -	L7805	Package_TO_SOT_THT:TO-220-3_Vertical

5. 2D PCB View screenshot



6. 3D PCB View screenshot



7. Bill of Materials (BOM)

Item	Qty	Reference	Value	Description	Footprint
1	1	C1	330nF	Ceramic disc capacitor (input filter)	Capacitor_THT:C_Disc_D5.0mm_W2.5mm_P5.00mm
2	2	C2, C4	10µF	Electrolytic capacitor	Capacitor_THT:CP_Radial_D5.0mm_P2.00mm
3	1	C3	100nF	Ceramic disc capacitor (output filter)	Capacitor_THT:C_Disc_D5.0mm_W2.5mm_P5.00mm
4	1	D1	LED	5 mm LED (power indicator)	LED_THT:LED_D5.0mm
5	1	J1	Barrel_Jack	DC power input jack	Connector_BarrelJack:BarrelJack_Horizontal
6	1	J2	5V_OUT	2-pin screw terminal (5V output)	TerminalBlock:TerminalBlock_Xinya_XY308-2.54-2P_1x02_P2.54mm_Horizontal
7	1	R1	1k	Resistor, LED current limiting	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_
8	1	U1	L7805	5V linear voltage regulator	Package_TO_SOT_THT:TO-220-3_Vertical